UNIT - 6
VISUALIZATION

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Contents

Data Visualization

Data visualization is the **graphical** or **visual format** representation of information and data. By using visual elements like **charts**, **graphs** and **maps**, data visualization tools provide an accessible way to see and understand **trends**, **outliers** and **patterns** in data. Additionally, it provides an excellent way for employees or business owners to present data to non-technical audiences without confusion.

Data visualization techniques such as **box plots, histograms, scatter plots** and **heat maps** help understand data patterns. **Tableau** enables interactive visualization, **outlier detection** identifies abnormal values and **data balancing** improves machine learning performance by handling imbalanced datasets.

Box Plot

A box plot displays distribution of numerical data, showing median, quartiles and outliers. It's useful for understanding variability and detecting unusual values.

A Box Plot (also called a **Box-and-Whisker Plot**) is a graphical method used in **data science and statistics** to visualize the **distribution, spread** and **variability of numerical data**.

It provides a summary of data using five important statistical values and helps in identifying **outliers, skewness** and **dispersion** in a dataset.

A box plot is especially useful when analyzing **large datasets** because it presents a compact summary of data distribution.

Box Plot

Real Life Example:

1. **Healthcare:** Detect abnormal patient reports.
2. **Finance:** Analyze stock prices and risks.
3. **Education:** Analyze student performance.
4. **Business:** Compare sales across regions.

Box Plot

A **Box Plot** is a statistical graph that displays data using:

1. Minimum value
2. First Quartile (Q1)
3. Median (Q2)
4. Third Quartile (Q3)
5. Maximum value

It also helps detect **outliers**.

Box Plot

1. **Minimum value:** The **smallest data value** excluding outliers.

   Example: Dataset: 5, 8, 10, 12, 15

   Minimum = **5**

2. **First Quartile (Q1):** The **middle value** of the dataset. It divides data into **two equal halves**. (50% of data lies below median)

3. **Median (Q2):** The **middle value** of the dataset. It divides data into **two equal halves**.

4. **Third Quartile (Q3):** Q3 represents the **75th percentile**. It separates the upper **25% of observations.**

5. **Maximum Value:** The **largest value** excluding outliers.

Mathematical Concept Behind Box Plot

Interquartile Range (IQR)

The spread of data is measured using:

$$IQR = Q3 - Q1$$

Outlier Detection Formula

Lower Bound: $Q1 - 1.5(IQR)$

Upper Bound: $Q3 + 1.5(IQR)$

Any value outside these limits is considered an **outlier**.

Mathematical Concept Behind Box Plot

Example

Dataset: 10, 12, 15, 18, 20, 22, 25, 28, 100.

Step 1: Find Quartiles

Q1 = 15

Median = 20

Q3 = 25

Step 2: Calculate IQR

IQR=25 − 15 = 10

Step 3: Find Limits

Lower Limit: 15 − 1.5(10) = 0

Upper Limit: 25 + 1.5(10) = 40

Since **100 > 40**, it is an **outlier**.

Advantages of Box Plot

Advantage	Explanation
Easy to understand	Simple graphical representation
Detects outliers	Identifies abnormal values quickly
Summarizes data	Shows distribution using five numbers
Compares datasets	Useful for multiple groups
Shows skewness	Indicates symmetry of data

Disadvantages of Box Plot

Advantage	Explanation
No exact values	Does not show actual data points
Less detail	Hides distribution shape
Not suitable for small data	Small datasets may mislead
No frequency info	Cannot show how many values exist
No exact values	Does not show actual data points

Histogram

A Histogram is a graphical representation used in **data science and statistics** to display the **frequency distribution of continuous numerical data**.

A Histogram is a graph that represents the distribution of numerical data by dividing data into intervals called **bins (or classes)** and showing the frequency of observations in each interval.

It helps in understanding:

- Data distribution
- Frequency of occurrence
- Shape of data
- Spread and variability
- Presence of skewness and outliers

Histogram

A histogram converts raw numerical data into visual form using **bars**. Unlike a bar chart, the bars in a histogram **touch each other**, indicating continuous data.

Real Life Example:

1. **Healthcare:** Age distribution of patients.
2. **Education:** Student marks distribution.
3. **Finance:** Income distribution.
4. **Marketing:** Customer purchase frequency.
5. **Machine Learning:** Feature analysis before model training.

Key Elements of Histogram

1. **Bins (Intervals/Class Intervals):** Bins are ranges into which data is divided.

   Example:

   Student marks:

   0–10

   10–20

   20–30

   30–40

Each interval is called a **bin**.

Key Elements of Histogram

2. Frequency: Frequency indicates **how many observations fall into a particular bin**.

Example

Marks Range	Frequency
0–10	5
10–20	8
20–30	12
30–40	6

Key Elements of Histogram

3. X-Axis:

Represents:

Data Intervals/Bins

Example:

- Age groups
- Salary range
- Marks range

4. Y-Axis

Represents:

Frequency of occurrence

Shows how often data values appear.

Key Elements of Histogram

4. Y-Axis

Represents:

Frequency of occurrence

Shows how often data values appear.

5. Bars

Bars represent frequencies.

Working of Histogram

Step 1: Collect Numerical Data

Example: Student marks:

12, 15, 18, 20, 22, 25, 30, 35

Step 2: Divide Data into Bins

Example:

Bin	Range
10–20	3
20–30	4
30–40	1

Working of Histogram

Step 3: Count Frequency

Count how many values fall in each interval.

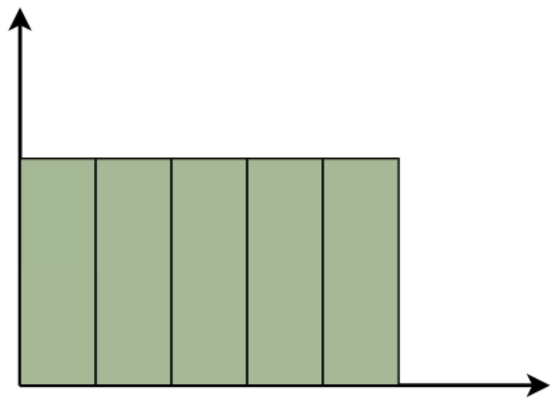
Step 4: Draw Bars

Height of each bar = frequency.

Types of Histogram Distribution

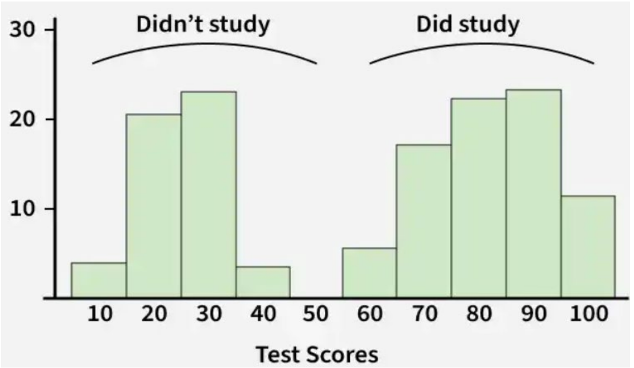
1. Uniform Histogram

A uniform histogram is a histogram in which all the bars are nearly equal in height, showing that the data is evenly distributed among all class intervals.



2. Bimodal Histogram

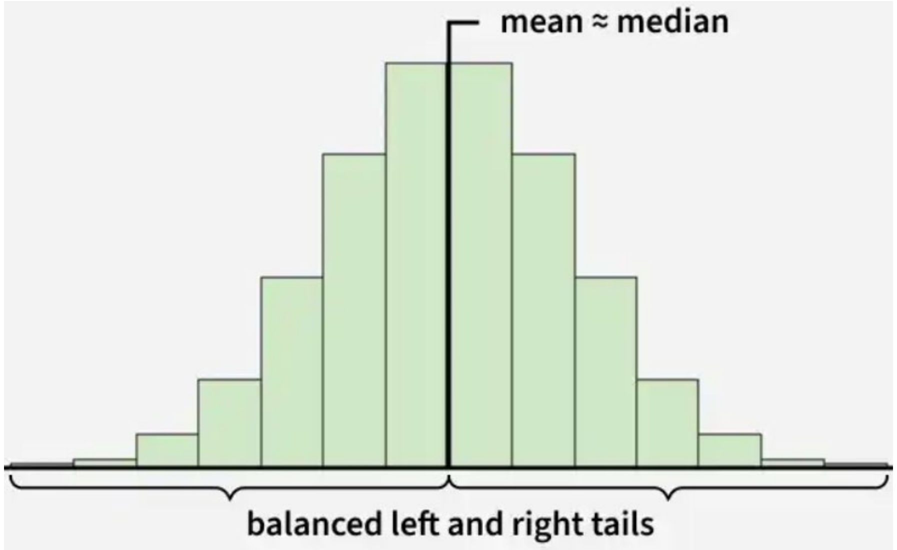
A histogram is called bimodal if it has two distinct peaks. This shows that the data consists of observations from two different groups or categories, and each group has its own centre or pattern of values.



Types of Histogram Distribution

3. Symmetric Histogram

A symmetric histogram is also called a bell-shaped histogram. It is said to be symmetric when a vertical line drawn through the centre divides the histogram into two identical halves. The left and right sides are equal in size and shape, showing a balanced distribution of data.

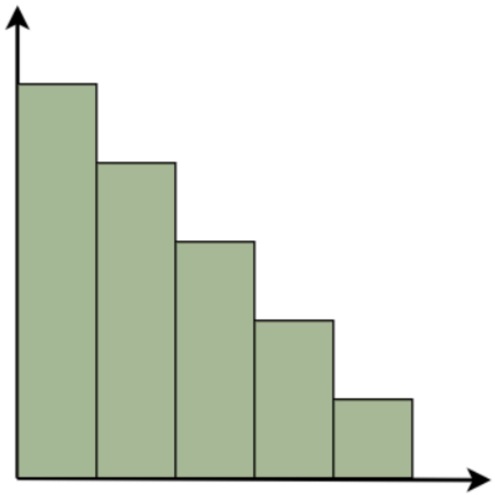


Types of Histogram Distribution

4. Right-Skewed Histogram

A right-skewed histogram is a histogram in which the bars extend more towards the right side. This means that most of the data values are on the left side, while a few very large values lie on the right.

For example: In a histogram showing family incomes, most families may have lower incomes, but a few very rich families stretch the data towards the right.

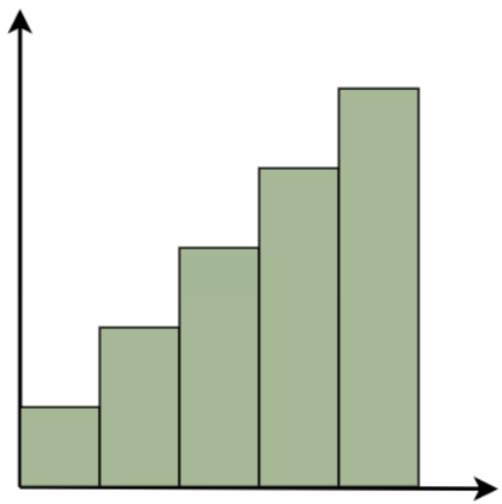


Types of Histogram Distribution

5. Left-Skewed Histogram

A left-skewed histogram shows bars that extend towards the left side. This means that most of the data values are on the right side, while a few very low values lie on the left.

For example: In a histogram showing test scores, most students may score high marks, but a few students with very low scores stretch the distribution towards the left.



Advantages of Histogram

Advantage	Explanation
Easy to understand	Simple visual representation
Shows distribution clearly	Displays frequency pattern
Detects skewness	Reveals symmetry
Useful for large datasets	Handles large numerical data
Helps detect outliers	Extreme values become visible

Disadvantages of Histogram

Disadvantage	Explanation
Bin size affects result	Wrong bins may mislead
No exact values shown	Only ranges visible
Hard to compare multiple datasets	Overlapping difficult
Not suitable for categorical data	Works only for continuous data

Scatter plot

A Scatter Plot is one of the most important **data visualization techniques** used in **data science, statistics, machine learning, and analytics** to study the **relationship between two numerical variables**.

It represents data points on a **two-dimensional graph**, where each point corresponds to one observation in the dataset.

Scatter plots are mainly used to:

- Identify **relationships (correlation)** between variables
- Detect **patterns and trends**
- Find **outliers**
- Analyze **clusters** in data

Scatter plot

A Scatter Plot is a graphical representation in which **individual data points are plotted on a coordinate plane** to show the relationship between **two continuous numerical variables**.

Each point in the graph represents one observation.

Real-Life example:

1. **Education:** Study hours vs marks
2. **Healthcare:** Exercise time vs blood pressure
3. **Business:** Advertising budget vs sales
4. **Finance:** Risk vs return
5. **Machine Learning:** Feature relationship analysis

Key Elements of Scatter Plot

1. X-Axis (Independent Variable)

The X-axis represents the **independent variable** or predictor variable.

Example:

- Study hours
- Age
- Advertising budget

Key Elements of Scatter Plot

2. Y-Axis (Dependent Variable)

The Y-axis represents the **dependent variable** whose value depends on X.

Example:

- Marks obtained
- Salary
- Sales amount

Example Table Study Hours Marks 2 40 4 55 6 70 8 90

Key Elements of Scatter Plot

3. Data Points (Dots)

Each dot represents **one observation** in the dataset.

Study Hours	Marks
2	40
4	55
6	70
8	90

Each row becomes one point on the graph.

Key Elements of Scatter Plot

4. Trend/Pattern

- Scatter plots reveal:
- Increasing trend
- Decreasing trend
- Random pattern

5. Correlation

Scatter plots help determine relationships between variables.

Types of Correlation in Scatter Plot

1. Positive Correlation

When both variables increase together.

Meaning: If X increases, Y also increases.

Example:

Study time ↑ → Marks ↑

2. Negative Correlation

When one variable increases and another decreases.

Meaning: If X increases, Y decreases.

Example:

Stress ↑ → Performance ↓

Types of Correlation in Scatter Plot

3. No Correlation

No visible relationship exists.

Example:

Shoe size vs intelligence

4. Strong Correlation

Points lie close to line.

5. Weak Correlation

Points spread widely.

Advantages of Scatter Plot

Advantage	Explanation
Easy visualization	Easy to understand relationships
Detects correlation	Shows positive/negative trends
Outlier detection	Easily identifies abnormal points
Useful for large datasets	Handles many observations
Supports prediction	Helpful in regression analysis

Disadvantages of Scatter Plot

Disadvantage	Explanation
Limited to two variables	Difficult for multiple variables
Overlapping points	Large data may overlap
Correlation ≠ Causation	Relationship doesn't prove cause
Hard with categorical data	Best for numerical data

Heat map

A Heat Map is a graphical representation of data where values are represented using **different colors**. It is one of the most powerful **data visualization techniques** used in **data science, machine learning, statistics** and **business analytics** to identify **patterns, relationships, trends** and **correlations** in datasets.

In a heat map, data values are displayed as **colors**, where:

1. **Dark/Hot colors** → Higher values
2. **Light/Cool colors** → Lower values

Heat maps help analysts quickly understand complex data by converting numbers into **color-coded visual patterns**.

Heat map

A Heat Map is a data visualization technique that represents numerical data using **color intensity**, where different colors indicate different magnitudes of values.

It helps in understanding:

1. Correlations
2. Trends
3. Concentrations
4. Relationships among variables

Real-Life Applications

1. **Healthcare:** Disease spread analysis.
2. **E-commerce:** Customer purchase analysis.
3. **Education:** Student performance visualization.
4. **Banking:** Fraud detection.
5. **Website Analytics:** User click tracking.

Working of Heat Map

Step 1: Collect Data

Data should be numerical.

Example: Website clicks:

Day	Clicks
Monday	120
Tuesday	200
Wednesday	80

Working of Heat Map

Step 2: Organize Data into Matrix

Rows and columns are created.

Step 3: Apply Color Intensity

Colors assigned according to value magnitude.

Higher value → darker color.

Step 4: Analyze Patterns

Observe:

- High concentration areas
- Correlations
- Trends

Types of Heat Maps

1. Correlation Heat Map

Used to show relationships between variables.

Example:

Feature correlation in machine learning.

Variables	Age	Salary	Experience
Age	1.0	0.8	0.7
Salary	0.8	1.0	0.9
Experience	0.7	0.9	1.0

Types of Heat Maps

2. Geographical Heat Map

Shows data intensity over geographic regions.

Example: COVID-19 spread map.

3. Website Heat Map

Tracks user activity.

Shows:

- Mouse clicks
- Scrolling behavior
- User attention areas

Types of Heat Maps

4. Cluster Heat Map

Groups similar patterns together.

Used in:

- Machine learning
- Data clustering

Uses of Heat Map in Data Science

Heat maps are widely used for:

1. Correlation Analysis

Understanding relationship between features.

2. Pattern Detection

Quickly identify trends.

3. Website Analytics

Analyze user clicks and behavior.

4. Machine Learning

Feature selection and correlation analysis.

Uses of Heat Map in Data Science

Heat maps are widely used for:

5. Business Intelligence

Sales and customer analysis.

6. Medical Analysis

Disease pattern visualization.

Advantages of Heat Map

Advantage	Explanation
Easy visualization	Converts numbers into colors
Quick pattern detection	Identifies trends instantly
Correlation analysis	Shows variable relationships
Handles large data	Useful for big datasets
Easy comparison	Compare multiple variables

Disadvantages of Heat Map

Disadvantage	Explanation
Hard for exact values	Colors show ranges only
Color dependency	Misleading color choice possible
Too much data clutter	Large matrices become confusing
Requires numerical data	Not useful for text/categorical data

Working with Tableau – Outlier detection

Tableau is a data visualization software that helps users **connect**, **analyze** and **visualize data** from multiple sources to generate meaningful insights.

Tableau is a powerful **Business Intelligence (BI)** and **data visualization tool** used in **data science, analytics** and **business intelligence** to **analyze, visualize** and **interpret data**.

It helps transform **raw data into interactive dashboards, charts, reports** and **visualizations** without requiring extensive programming knowledge.

Tableau supports:

1. Data analysis
2. Dashboard creation
3. Reporting
4. Data exploration
5. Predictive analytics

Features of Tableau

1. Drag-and-Drop Interface

Users can create visualizations without coding.

2. Interactive Dashboards

Allows dynamic filtering and interaction.

3. Real-Time Data Analysis

Supports live data connections.

4. Multiple Data Source Integration

Can connect with: **Excel, CSV files, SQL databases, Google Sheets, Cloud databases**

5. Big Data Support

Can process very large datasets.

Applications of Tableau

1. Healthcare
2. Banking
3. Education
4. Marketing
5. Business analytics
6. Data science

Types of Visualizations in Tableau

Tableau supports:

Visualization	Purpose
Bar Chart	Comparison
Histogram	Distribution
Scatter Plot	Relationship
Heat Map	Correlation
Pie Chart	Percentage
Line Chart	Trends

Advantages of Tableau

Advantage	Explanation
Easy to use	Drag-and-drop interface
Interactive dashboards	Dynamic visualization
Fast processing	Quick analysis
Multiple data sources	Flexible integration
Real-time analytics	Live updates

Disadvantages of Tableau

Disadvantage	Explanation
Expensive license	Paid software
Limited customization	Compared to coding
Large data complexity	Performance issues

Outlier Detection in Data Science (Detailed Explanation)

Introduction:

In data science, Outlier Detection is the process of identifying **unusual, abnormal** or **extreme data points** that significantly differ from the rest of the dataset.

Outliers may occur because of:

- Data entry errors
- Experimental errors
- Fraudulent activities
- Rare events

Outlier Definition:

An Outlier is a data point that **lies significantly far away from other observations** in a dataset.

Example:

Dataset: 10, 12, 15, 18, 20, 22, 25, 150.

Here, **150** is an outlier.

Applications

1. Fraud detection
2. Medical diagnosis
3. Cybersecurity
4. Finance
5. Machine learning

Why Outlier Detection is Important

Outliers can:

- Distort data analysis
- Reduce model accuracy
- Affect machine learning performance

Hence, detecting them is important.

Types of Outliers

1. **Global Outlier:** Very different from entire dataset.

   Example: Salary = ₹10 lakh among ₹20,000 salaries.

2. **Contextual Outlier:** Abnormal in specific situation.

   Example: 30°C temperature in winter.

3. **Collective Outlier:** Group of unusual observations.

   Example: Suspicious bank transactions.

Methods of Outlier Detection

1. Box Plot Method (IQR Method)

Most commonly used.

Formula:

$$IQR = Q3 - Q1$$

Lower Limit:

$$Q1 - 1.5(IQR)$$

Upper Limit:

$$Q3 + 1.5(IQR)$$

Values outside limits are outliers.

Methods of Outlier Detection

1. **Box Plot Method (IQR Method)**

Example

Dataset: 10,12,15,18,20,22,25,100

Here, 100 becomes outlier.

2. Z-Score Method

Measures distance from mean.

Formula:

$Z = (X - \mu) / \sigma$ If: $|Z| > 3$

Then point is an outlier.

Methods of Outlier Detection

3. Scatter Plot Method

Outliers appear far away from clusters.

4. Histogram Method

Extreme values become visible.

5. Machine Learning Methods

Used in large datasets:

1. Isolation Forest
2. DBSCAN
3. Local Outlier Factor (LOF)

Handling Outliers

1. **Remove Outliers:** Delete abnormal values.
2. **Replace Values:** Use:
   - Mean
   - Median
3. **Transformation:** Apply:
   - Log transformation
   - Scaling
4. **Keep Outliers:** Useful if meaningful.

   Example: Fraud detection.

Advantages of Outlier Detection

Advantage	Explanation
Improves accuracy	Better ML performance
Detects fraud	Banking/security
Better analysis	Removes noise

Disadvantages of Outlier Detection

Disadvantage	Explanation
Risk of data loss	Useful values removed
Computational cost	Large data complexity
False detection	May remove important data

Data Balancing

In **data science and machine learning**, Data Balancing is the process of adjusting an **imbalanced dataset** so that all classes are represented equally or fairly.

An imbalanced dataset occurs when one class contains **significantly more observations** than another class.

For example:

Class	Count
Yes	950
No	50

Here, the dataset is **imbalanced** because one class dominates.

Data balancing helps machine learning models learn equally from all classes and improves prediction performance.

Data Balancing

Definition

Data Balancing is the technique of modifying a dataset to ensure that all target classes have a balanced or nearly balanced number of observations.

Its goal is to **reduce bias and improve model performance**.

Why Data Balancing is Important

In machine learning, imbalanced data causes models to favor the **majority class** and **ignore the minority class**.

Example: Fraud Detection Dataset:

Transaction Type	Count
Genuine	99,000
Fraud	1,000

Without balancing:

- Model predicts everything as genuine.
- Fraud cases remain undetected.

Importance of Data Balancing

Data balancing helps in:

1. Improving prediction accuracy
2. Preventing biased models
3. Better minority class prediction
4. Enhancing model reliability
5. Improving classification performance

Imbalanced Dataset Problem

An imbalanced dataset can cause:

1. **Biased Model:** Model favors majority class.

2. **Poor Accuracy for Minority Class:** Minority predictions become weak.

3. **Overfitting:** Model memorizes dominant class.

4. **Misleading Accuracy**

   Example: If 95% data is class A:

   Model predicting only A gives 95% accuracy but performs poorly.

Types of Imbalanced Data

1. **Binary Class Imbalance:** Only two classes exist.

    Example: Spam / Not Spam

2. **Multi-Class Imbalance:** Multiple classes are uneven.

    Example:

Grade	Count
A	100
B	500
C	50

Methods of Data Balancing

There are mainly three approaches:

A. Oversampling

Definition: Oversampling increases the number of observations in the **minority class**.
It duplicates or creates synthetic samples.

Example: Before Balancing:

Class	Count
Yes	900
No	**100**

After Oversampling:

Class	Count
Yes	900
No	**900**

Methods of Data Balancing

There are mainly three approaches:

Types of Oversampling:

1. **Random Oversampling:** Duplicates minority data randomly.

   Example: Copying minority records.

2. **SMOTE (Synthetic Minority Over-sampling Technique):** Generates **synthetic/artificial data points** instead of duplication.

   Working of SMOTE

   1. Select minority sample
   2. Find nearest neighbors
   3. Generate synthetic values

Advantages of Oversampling

Advantage	Explanation
Prevents data loss	Keeps all majority data
Better minority prediction	Improves learning
Improves accuracy	Better classification

Disadvantages of Oversampling

Disadvantage	Explanation
Overfitting risk	Duplicate data problem
Increased training time	More data created

B. Under sampling

Definition: Under sampling reduces the number of observations in the majority class.

Example

Before Balancing:

Yes	**900**
No	100

After Under sampling:

Yes	**100**
No	100

Types of Under sampling:

1. **Random Under sampling:** Randomly removes majority data.
2. **Cluster Centroid:** Uses clustering to reduce majority samples.

Advantages of Under sampling

Advantage	Explanation
Faster training	Less data
Reduces complexity	Small dataset

Disadvantages of Under sampling

Disadvantage	Explanation
Data loss	Useful information removed
Lower accuracy	Important patterns lost

C. Hybrid Sampling

Definition: Combination of Oversampling and Under sampling

Example: SMOTE + Under sampling

Advantage

- Balances performance and efficiency.

Data Balancing Workflow

Raw Dataset

↓

Check Class Distribution

↓

Imbalanced?

↓

Choose Method

(Over/Under/Hybrid)

↓

Balanced Dataset

↓

Train ML Model

Evaluation Metrics for Imbalanced Data

Accuracy alone is misleading.

Important metrics:

1. Precision

$$\text{Precision} = TP / (TP + FP)$$

2. Recall

$$\text{Recall} = TP / (TP+FN)$$

3. F1 Score

$$\text{F1Score} = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$$

4. ROC-AUC

Measures classification quality.

Applications of Data Balancing

1. **Fraud Detection:** Credit card fraud.

2. **Medical Diagnosis:** Rare disease prediction.

3. **Spam Detection:** Spam vs non-spam emails.

4. **Cyber security:** Intrusion detection.

5. **Customer Churn Prediction:** Identifying leaving customers.

Advantages of Data Balancing

Advantage	Explanation
Improves prediction	Better minority classification
Reduces bias	Fair learning
Better ML performance	Improved accuracy
Reliable results	Balanced learning

Disadvantages of Data Balancing

Disadvantage	Explanation
Overfitting risk	Oversampling issue
Data loss	Undersampling issue
Increased complexity	Advanced methods

Data Balancing conclusion

Data Balancing is an essential preprocessing technique in data science that helps solve the problem of **imbalanced datasets**. By using techniques such as **over sampling, under sampling** and **SMOTE**, balanced datasets improve machine learning performance, reduce bias, and ensure accurate prediction for minority classes.

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